

Chapter 132 — Surgical Management of Bladder Cancer: Transurethral, Open and Robotic

INTRODUCTION

Epidemiology and Framing

- Bladder cancer is **one of the commonest non-cutaneous malignancies** in the United States; the American Cancer Society projected roughly 82,000 new cases and 16,700 deaths for 2023.
- Without appropriate surgery the disease is **lethal and carries high health-care cost**; surgery for both non-muscle-invasive bladder cancer (NMIBC) and muscle-invasive bladder cancer (MIBC) demands attention to technique, oncological principle and disease pathogenesis.
- **Transurethral resection (TUR)** establishes the pathological diagnosis and local extent, and is itself a **complete oncological procedure for low-grade NMIBC**.
- **MIBC requires radical cystectomy with regional lymph node dissection** for local control and the best chance of cure.

History

MILESTONES IN BLADDER CANCER SURGERY

Date	Figure	Contribution
1887	Bardenheuer (Germany)	First cystectomy , for advanced tumour; no urinary diversion; patient died of renal failure
Same era	Nitze (urologist) and Leiter (instrument maker)	First modern endoscope , forerunner of current instruments
1956	Paquin and Marshall	First modern radical cystectomy technique : in men began with perineal exploration and dissection of rectum off bladder and prostate under vision; favoured cutaneous ureterostomy
Early series	Whitmore and Marshall	Substantial morbidity; limited one-year and five-year survival

- Refinement since has produced **low contemporary inpatient and 90-day mortality** and a steady rise in five-year survival in large high-volume series.

Neoadjuvant Chemotherapy

- **Neoadjuvant cisplatin-based chemotherapy improves survival** — landmark randomised trial (Grossman, 2003): **median survival 46 → 77 months**. It should be **considered in every eligible MIBC patient** scheduled for radical cystectomy.

PROGRESSED VERSUS DE NOVO MUSCLE-INVASIVE BLADDER CANCER

	Progressed MIBC (P-MIBC)	De novo MIBC (D-MIBC)
Definition	NMIBC treated with bacille Calmette–Guérin (BCG) that progresses to MIBC	MIBC at first presentation
Historical prognosis	Worse — but in studies predating routine neoadjuvant chemotherapy (NAC)	Better

	Progressed MIBC (P-MIBC)	De novo MIBC (D-MIBC)
Response to cisplatin NAC	One modern study suggests a molecular basis for chemoresistance ; others find no difference in pathological response or outcome after NAC and radical cystectomy (RC)	Reference group

- The clinicopathological and molecular diversity of these patients means a **uniform approach is insufficient**; whether BCG-treated progressors should receive NAC remains controversial.

Evolving Systemic Therapy

- Perioperative trials are testing **immune checkpoint inhibitors with or without chemotherapy and antibody-drug conjugates**, with correlative and biomarker secondary endpoints intended to refine patient selection; the treatment paradigm, including early-stage disease, is expected to change.

Concept — *Stage decides the operation: TUR is diagnosis for everyone and cure for low-grade NMIBC, whereas MIBC needs radical cystectomy with lymphadenectomy, and modern survival gains owe as much to cisplatin-based neoadjuvant therapy as to surgical refinement.*

TRANSURETHRAL RESECTION OF BLADDER TUMOURS

Preoperative Evaluation

- **History directed at risk factors:** tobacco; drugs — cyclophosphamide, phenacetin; occupational and environmental carcinogens — aromatic amines, polycyclic aromatic hydrocarbons, trihalomethanes.
- **Surgical implants and valvular heart disease** alter antibiotic prophylaxis; check against current guidelines.
- **Bleeding history and anticoagulants:** aspirin is generally continued (no significant bleeding observed); caution with warfarin, heparin, rivaroxaban, apixaban, clopidogrel, ticagrelor and prasugrel — stopping time set by each drug's half-life, and bridging agreed preoperatively with the relevant team.
- Examination: **suprapubic palpation** for a mass; **digital rectal examination (DRE)** in men if prostatic involvement is suspected; **bimanual examination in women** for anterior vaginal involvement.
- Laboratory: full blood count, metabolic panel, coagulation panel, **urinalysis with culture; active infection → defer resection until bacteria are cleared.**
- **Upper tract imaging completes the diagnostic work-up:** renal ultrasound with retrograde pyelogram, computed tomography (CT) urogram (CTU), magnetic resonance imaging (MRI), and rarely intravenous pyelogram.

CONTRAST AGENTS AND RENAL FUNCTION

Agent	Permitted	Contraindicated
Iodinated intravenous contrast (CT)	Glomerular filtration rate (GFR) ≥60 mL/min	Below this threshold — use an alternative
MRI contrast (gadolinium-based)	GFR >30 mL/min	GFR <30 mL/min

- **MIBC → formal staging of chest, abdomen and pelvis.**
- Office flexible cystoscopy usually precedes TUR: confirms a mucosal tumour and **records location and extent** for planning.
- Experienced clinicians can predict stage and grade on follow-up cystoscopy, but **tissue must be obtained at baseline even when low grade is suspected.**

- **Tumour over a ureteric orifice** → be prepared for ureteroscopy, ureteric stent and fluoroscopy.

VALUE OF MAXIMAL RESECTION BY STAGE

NMIBC	MIBC proceeding to radical cystectomy (with or without NAC)
Complete eradication of visible disease improves oncological outcome	No additional survival benefit from maximal TUR of bladder tumour (TURBT); residual disease on TURBT predicts overall and recurrence-free survival but removing it does not improve them
—	Clinical staging is unreliable: a substantial proportion of cT2 tumours are extravesical at cystectomy

Operative Set-up and Inspection

- **Sequence:** rigid cystoscope, **dorsal lithotomy** → **antibiotic prophylaxis within 1 hour** → **30° lens with visual obturator** (limits urethral trauma): whole urethra, preliminary bladder survey, ureteric orifices → **70° lens:** complete survey, especially **bladder neck, dome and anterior wall** → **120° lens** if still incomplete → record location and extent → **blue-light cystoscopy** where available for lesions poorly seen under white light, such as carcinoma in situ (CIS; see Chapter 130).
- Antibiotic prophylaxis is **recommended by the American Urological Association (AUA)** — extrapolated from a randomised trial in TUR of the prostate showing less postoperative bacteriuria.
- **Avoid fluoroquinolones where possible:** the US Food and Drug Administration (FDA) label change of **July 2016** cautioned against their use in uncomplicated urinary tract infection (no specific statement on prophylaxis).
- **AUA and Society of Urologic Oncology (SUO) joint guideline: enhanced cystoscopy — narrow-band imaging and blue-light cystoscopy — if available;** both increase tumour detection and reduce recurrence.

ENERGY SOURCE AND IRRIGANT

	Monopolar	Bipolar
Irrigant	Sterile water or 1.5% glycine	0.9% sodium chloride
Postoperative electrolyte disturbance	Higher risk	Lower risk
Obturator reflex (lateral wall)	Present	Mitigated

Resection Principles

- Goal: **visual eradication of all tumour and an adequate depth** — loop electrode, cutting current, **specimen to include muscularis propria** (Fig. 132.1).
- Tumour extends histologically beyond its visible edge → include a **margin of approximately 2 cm of normal-appearing tissue** where possible; wide resection secures completeness for both **broad-based and tentacular** growth patterns (Fig. 132.2).
- After visual completion: empty the bladder → **bimanual examination under anaesthesia; a mass still palpable after resection is staged clinical T3b (cT3b).**

Specific Challenges

DIFFICULT TUMOUR LOCATIONS

Site	Hazard	Countermeasures
Lateral wall	Obturator reflex: current stimulates obturator nerve → ipsilateral leg adduction → lateral deflection of instrument → perforation	Minimal bladder distension; bipolar energy; muscle relaxant under general anaesthesia
Ureteric orifice	Cauterising the orifice closed → stricture, hydronephrosis	Pure cutting current, quick strokes; routine stenting not necessary (benign orifice stricture uncommon in a retrospective series); authors would place a short-term stent after large resection or extensive cautery, though ultimate stricture rate is probably unchanged
Dome	Angulation, distance, intraperitoneal perforation; heat transmission to bowel even without perforation	Minimal distension; suprapubic pressure (non-dominant hand or assistant); straight-ahead loop; avoid prolonged cautery

Bladder Perforation

- Perforation may be **the natural result of an adequate resection**, especially of advanced tumours (Fig. 132.3).

EXTRAPERITONEAL VERSUS INTRAPERITONEAL PERFORATION

	Extraperitoneal (Fig. 132.3A)	Intraperitoneal (Fig. 132.3B)
Typical site	Resections away from posterior wall and dome	Posterior wall and dome
Frequency and severity	Commoner	Rare, but large ones cause significant morbidity
Intraoperative clues	—	Loss of bladder distension, visible posterior or dome defect, palpable abdominal distension
Confirmation	Cystogram at the same sitting	Cystogram at the same sitting
Management	Urethral (Foley) catheter drainage and observation	Laparotomy, meticulous bowel inspection, repair; catheter and abdominal drain

Adjuvant Intravesical Therapy

- Intravesical therapy after TUR **reduces recurrence with little or no effect on progression.**

POSTOPERATIVE INTRAVESICAL AGENTS

Agent	Setting	Timing and dose	Bottom line
Single-dose chemotherapy (mitomycin C, epirubicin)	NMIBC	Within 24 hours of TUR	Meta-analysis of randomised trials: about 13% absolute reduction in recurrence
Gemcitabine — Southwest Oncology Group (SWOG) S0337 (landmark)	Suspected low-grade NMIBC	Immediately after TUR, 2 g in 100 mL saline vs saline	Well tolerated, no grade 4–5 toxicity; 12% absolute reduction in recurrence at 4 years (hazard ratio [HR] 0.66, P <0.01)

Agent	Setting	Timing and dose	Bottom line
BCG	High-grade NMIBC — most active agent	Never immediately: wait 2–4 weeks and confirm no gross haematuria	—

- **Adjuvant instillation only in the absence of perforation.**

Concept — TURBT serves two purposes by stage: in NMIBC completeness and depth are therapeutic, in MIBC they are only prognostic and staging information; every technical choice (energy, distension, cutting current) aims to put detrusor in the specimen without perforating, because perforation also forfeits immediate intravesical chemotherapy.

PATIENT PREPARATION

Staging

- **CT of chest, abdomen and pelvis** if renal function is adequate and there is no iodinated contrast allergy; **MRI substitutes when renal function is impaired.**
- Contrast cross-sectional imaging assesses the upper tracts and regional and distant spread, and **guides perioperative chemotherapy.**
- MRI, with better soft-tissue resolution, assesses depth of invasion and **can help distinguish NMIBC from MIBC before TUR.**
- Nodal staging: a systematic review suggests **MRI and positron emission tomography–CT (PET/CT) may be more sensitive than CT**, yet **CT remains the commonest modality** (cost, availability, speed); **CT nodal accuracy varies widely with the node-size threshold.**
- **Bone and brain imaging only when clinically indicated.**

Medical Optimisation and Stoma Marking

- Expect **coronary artery disease, smoking-related lung disease and peripheral vascular disease.**
- **Enterostomal therapist stoma marking for every patient, whatever the planned diversion** — conversion to an ileal conduit during a planned continent diversion is rare but must be consented.
- Mark **standing and seated** to avoid folds and maximise appliance fit; without a stomal therapist the **surgeon marks, patient awake and in usual clothing.**

Bowel Preparation

- Mechanical preparation was intended to reduce anastomotic leak and wound and abdominal infection; **randomised trials in colorectal surgery showed no benefit** — complications, leak, wound infection and abscess were similar without it.
- In radical cystectomy, pooled randomised trials show **no or limited preparation → faster return of bowel activity and flatus, less fever and fewer wound complications**, without other harm.
- **Routine mechanical bowel preparation is not advised**, especially when only ileum is used; **no routine oral antibiotic adjunct.**

Alvimopan

- **Peripherally acting μ -opioid receptor antagonist**, given **30–90 minutes before surgery** and twice daily thereafter.
- Phase III trial (bowel resection): **less postoperative ileus, earlier discharge**; phase IV study (radical cystectomy): **shorter stay, less total parenteral nutrition, cost saving.**

Antibiotic Prophylaxis

- **Intravenous, within 1 hour of incision**, tailored to local susceptibility; cover **gram-positive skin flora** and **gram-negative aerobes and anaerobes** of distal small and large bowel — a **broad-spectrum cephalosporin** such as **cefoxitin** is usually adequate.

Venous Thromboembolism Prophylaxis

- High-risk patients without significant bleeding: **mechanical (pneumatic compression) plus pharmacological prophylaxis, started before induction** of general or spinal anaesthesia.
- **Extended prophylaxis for 4 weeks**: a randomised trial of 28 vs 8 days after abdominal or pelvic surgery showed a **marked reduction in major venous thromboembolism (VTE) without more bleeding**.

AGENTS FOR EXTENDED VTE PROPHYLAXIS

	Enoxaparin (low-molecular-weight heparin, LMWH)	Direct oral anticoagulant (DOAC), e.g. apixaban
Status	Most used after cystectomy	Safe, effective alternative
Compliance	Poorer: cost, self-injection, injection-site complications	Better (oral)
Evidence	Reference	Cancer-associated VTE meta-analysis: no difference in recurrent VTE, major or clinically relevant non-major bleeding; after urological cancer surgery, apixaban → fewer major complications, better compliance
Verdict	—	Consider DOAC for extended prophylaxis after radical cystectomy

Positioning and Skin Preparation

- **Men**: supine, table flexion point at the anterior superior iliac spine, flexed about **15°** (less after spinal fusion or lumbar injury) (Fig. 132.4A).
- **Women**: **low lithotomy** (stirrups or spreader bars) for vaginal access; table flexion generally not possible.
- Field: xiphoid to upper thighs, including genitalia, vagina and perineum.
- **10% povidone-iodine; avoid chlorhexidine gluconate on genital skin.**

Concept — Preparation is enhanced recovery by subtraction and addition: remove what does not help (bowel preparation, oral antibiotics) and add what protects bowel recovery and prevents clot (alvimopan, a timed intravenous antibiotic, four weeks of thromboprophylaxis).

SURGICAL TECHNIQUE

Exposure and Bladder Mobilisation

- **Sequence**: lower midline incision, **pubic symphysis → umbilicus** (infraumbilical often suffices; extend cephalad as needed) (Fig. 132.4B) → **fascia divided exactly in the midline** (aids closure, avoids detaching rectus abdominis from its pubic insertion); upward umbilical traction reveals the linea alba → **space of Retzius** entered; blunt release of bladder from pelvic sidewalls cephalad to **vas deferens (men) / round ligament (women)** → peritoneotomy lateral to a medial umbilical ligament; **urachus controlled and divided** (Fig. 132.5) → peritoneum incised lateral to both medial umbilical ligaments to the internal inguinal rings; vasa deferentia or round ligaments divided.

Bowel Mobilisation

- **Right:** white line of Toldt incised around the caecum, posterior peritoneum opened to **mobilise the root of the small bowel mesentery**.
- **Left:** white line of Toldt incised and a **window created beneath the sigmoid mesentery**, joining the right posterior peritoneotomy — the **route for transposing the left ureter to the right lower quadrant** (Fig. 132.6).
- Bookwalter-type retractor, bowel packed cephalad; **confirm with the anaesthetist that the vena cava is not compressed**; moist packs behind retractor blades.

Ureteric Dissection

- **Sequence:** free each ureter from a few centimetres above the iliac crossing **to the detrusor hiatus** → **preserve adventitia** → **ligate and divide the obliterated umbilical artery first** (maximises ureteric length) → ligate (tie or suture ligature) and divide.
- **Distal ureteric margin frozen section is controversial:** a positive margin correlates with later upper tract recurrence, but a survival effect is unproven, and **sequential resection to a negative margin does not eliminate upper tract recurrence**.
- Urine management by preference: **temporary ureteric catheters led off the field**, or temporary ligation to avoid spillage.

Concept — *Ureteric length and blood supply are the currency of the later diversion: adventitia is preserved, the obliterated umbilical artery divided early, and the left ureter given a ready-made retrosigmoid tunnel.*

PELVIC LYMPHADENECTOMY

- Performed **before or after removal of the bladder**, by surgeon preference.

LYMPHADENECTOMY TEMPLATES

Template	Boundaries / content	Evidence
Standard (Fig. 132.7A)	Lateral: genitofemoral nerve; medial: internal iliac artery; distal: Cooper ligament; proximal: ureter crossing common iliac artery	Reference template
Extended (Fig. 132.7B — node packets)	Adds entire common iliac and presacral packets ; used in advanced disease	Better staging; possible oncological benefit over restricted templates
Superextended	Further to preaortic packet at inferior mesenteric artery	No extra staging information beyond the common iliac level; no benefit over extended

- Technique: **protect the obturator nerve**, control lymphatics at caudal and cephalad extremes, confirm no residual lymphoid tissue (Fig. 132.7C–D).
- **Node yield is a surrogate of surgical quality:** **≥10 nodes removed → better five-year survival**; in node-positive disease (Surveillance, Epidemiology, and End Results population data) **≥10 nodes vs 1–5 → longer overall survival**.
- Extent: systematic review — extended or superextended may outperform restricted templates, superextended adds nothing over extended; **randomised trial: extended not superior to limited** for recurrence-free or cancer-specific survival.
- Therapeutic value of extended dissection remains controversial, but **its staging value is secure** — it identifies candidates for adjuvant systemic therapy; mapping shows a minority of node-positive patients have **no true-pelvis metastases**, most of whom only extended dissection would detect.

- **NMIBC at radical cystectomy:** meaningful upstaging to $\geq pT2$ and occult node positivity; **yield >10 → better local pelvic recurrence-free survival; yield >20 → better cancer-specific and overall survival**, most evident in cTis and cT1.
- Completed dissection exposes the **vascular pedicles** (Fig. 132.8): **superior, middle and inferior vesical arteries** controlled with vascular stapler, clips or sealing device.

Concept — *Lymphadenectomy is certain as staging and uncertain as therapy: yield thresholds track quality and survival, but the randomised comparison did not show extended dissection beat limited dissection.*

RADICAL CYSTECTOMY: MALE

Posterior Dissection

- **Sequence:** anterior pedicles ligated → rectal cul-de-sac identified; **peritoneum over the seminal vesicles incised** (Fig. 132.9) — ensure margin if tumour at bladder base → rectum freed in midline (blunt or sharp) to the prostate → **Denonvilliers fascia incised** → dissection carried laterally; **posterior pedicles** identified (Fig. 132.10A) and controlled with clips, ties, staplers or sealers (Fig. 132.10B).
- **Advanced disease, prior pelvic radiation, or fibrosis after resection or intravesical chemotherapy → no blunt dissection; sharp dissection under vision.**
- **Rectal injury → primary repair**, with or without flap coverage and/or faecal diversion.
- **Sealing devices transmit heat to the rectum → shield the rectum with a gloved finger** from the instrument tips.

Anterior and Apical Dissection

- **Sequence:** urethra now palpable → **endopelvic fascia** incised over levators → urethra–dorsal venous complex confluence identified → **dorsal venous complex ligated and divided** (Fig. 132.11) → anterior urethra incised.

Urethral Margin and Neobladder

- **Planned orthotopic neobladder → preserve urethral length and send a urethral margin frozen section.**
- Urethral recurrence occurs in a minority, usually within two years; **risk is lower after orthotopic than cutaneous diversion** (reason unclear).
- **Negative intraoperative frozen section → no urethral recurrence at 10 years** in one series; **preoperative prostatic urethral biopsy correlates poorly** with final margin, whereas a **negative frozen section has very high negative predictive value.**
- **Positive urethral margin → no orthotopic neobladder;** warn the patient preoperatively.

Nerve Sparing

- **Controversial, unlike in radical prostatectomy;** technique is analogous (Fig. 132.12).
- Functional results are **significantly worse than after prostatectomy and not clearly better than non-nerve-sparing cystectomy;** only a minority regain erections adequate for intercourse at one year, more after neobladder than ileal conduit.
- **Age is a strong predictor** of recovery, as after prostatectomy.
- Local recurrence after nerve sparing is uncommon but **has occurred even in pT2 disease.**
- Decide by preoperative sexual function and disease burden; **non-nerve-sparing (Fig. 132.13) is preferred in advanced disease and pre-existing erectile dysfunction.**

Prostate Sparing (Subtotal Prostatic Resection)

- Variants: whole prostate left, or prostatic capsule and/or seminal vesicles spared; studied mainly with **orthotopic neobladder**.
- Early series: excellent erectile function and no local recurrence, but **short follow-up**; meta-analysis: **better potency and better day- and night-time continence**.
- Concern: **high rate of occult prostate cancer in cystoprostatectomy specimens**; in carefully selected men (no prostate cancer preoperatively, negative frozen section) local control is good, **but distant metastasis remains frequent**.
- **Work-up: DRE, prostate-specific antigen (PSA), transurethral prostatic resection, intraoperative frozen section; abnormal PSA or DRE → prostate biopsy before surgery.**

Concept — Every functional preservation in men is paid for at an oncological margin: the urethra for a neobladder (the frozen section decides), the nerves for potency (a disappointing yield), and the prostate for erection and continence (the occult prostate cancer risk).

RADICAL CYSTECTOMY: FEMALE

Rationale and Counselling

- Traditional operation: **anterior pelvic exenteration** — bladder, urethra, anterior vagina, uterus, cervix; essential with a positive examination under anaesthesia or suspected **anterior vaginal wall invasion (cT4a)**.
- Women **present with more advanced disease**, and **female sex is an independent risk factor for death from disease** → **anterior exenteration remains the gold standard**.
- Sexual morbidity: **limited recovery of sexual function by two years**; dyspareunia, reduced libido and vaginal dryness common; more than half receive no preoperative counselling → **counsel thoroughly**.
- **Pelvic organ sparing** can be considered in highly selected women prioritising sexual function: gynaecological organ involvement is uncommon in MIBC and ovarian involvement was absent in one series.
- **The ovaries continue to produce testosterone after menopause**; bilateral salpingo-oophorectomy impairs sexual function → **discuss ovarian sparing** when disease is not locally advanced.
- **cT1–cT2 with orthotopic neobladder planned** → **vaginal and urethral sparing are necessary**.

Anterior Exenteration Technique

- **Sequence:** bowel and anterior bladder mobilisation and ureteric dissection as in men, except **ovarian vessels ligated (2-0 silk distally; 2-0 silk suture ligature plus tie proximally) and divided** → identify **posterior cervical fornix** (Fig. 132.14A) → **incise vaginal cuff there** (Fig. 132.14B) → control lateral and posterior pedicles (staplers, sealers or clips) → specimen: uterus, cervix, anterior vaginal cuff, bladder → **urethral meatus incised** antegrade or from the introitus; specimen removed (Fig. 132.15A–B) → keep **enough vaginal mucosa above the meatus** for closure → release posterior vaginal wall from rectum if needed (Fig. 132.15C) → **clamshell closure** of posterior flap to introitus with 2-0 polyglactin 910 (**preserves girth at the cost of length**) → **watertight, interrupted closure** → **vaginal pack, removed on day 2**.
- The female pelvis is vascular and the pedicles are **sinusoidal over the lateral vaginal wall** → meticulous haemostasis.
- Non-watertight closure → **troublesome drainage of peritoneal fluid**; the pack distends the vagina, tamponades wall bleeding (particularly after vaginal sparing) and reveals defects.

Vaginal- and Urethral-Sparing Cystectomy for Neobladder

- **Orthotopic neobladder** if the bladder neck is uninvolved and disease is $\leq$ cT2.
- Requires **urethral length proximal to the striated sphincter** and **anterior vaginal wall sparing** to support the neobladder.
- **Sequence:** separate lateral pedicles from lateral vaginal wall before ligation — after removing cervix and uterus at the fornix (Fig. 132.16A) or with them in place → **vaginal pack or end-to-end anastomosis (EEA)** sizer defines the midline bladder–vagina plane → extend laterally → **do not divide the pedicles until the midpoint of the lateral vaginal wall is reached** (preserves bladder margin) → continue to **bladder neck (Foley balloon as guide)** → preserve striated sphincter; transect → **urethral margin frozen section; malignancy → orthotopic diversion contraindicated** → vaginal apex closed with 2-0 polyglactin 910; urethral anastomotic sutures placed (Fig. 132.16B–C).

ORGAN- AND FUNCTION-SPARING OPTIONS

Option	Candidate	Gain	Price / precondition
Male nerve sparing	Good preoperative function, limited disease	Modest potency benefit	Poor functional yield; local recurrence reported even in pT2
Male prostate sparing	Normal DRE and PSA, negative transurethral resection and frozen section	Potency; day and night continence	Occult prostate cancer; distant metastasis
Female ovarian sparing	Not locally advanced	Postmenopausal testosterone, sexual function	Individual risk–benefit discussion
Female vaginal and urethral sparing	$\leq$ cT2, bladder neck uninvolved	Orthotopic neobladder; sexual function	Negative urethral frozen section

Concept — In women the default operation is wider than in men because they present later and die more often of the disease; sparing is the exception, earned by low stage, an uninvolved bladder neck and a negative urethral margin.

PARTIAL CYSTECTOMY

Selection and Outcomes

- **Solitary, small MIBC without concurrent CIS** → results similar to radical cystectomy; population-based cohort matched for node count and clinicopathology: **equivalent overall and cancer-specific survival**.
- Recurrence pathways: **NMIBC recurrence** → intravesical therapy; **muscle-invasive recurrence** → **salvage radical cystectomy**, but survival after salvage is far worse if disease is extravesical or node-positive than if organ-confined.
- Preoperative: **counsel about possible radical cystectomy and diversion** (margin may be unattainable or extravesical disease found); **cisplatin-eligible** → offer neoadjuvant chemotherapy.

Technique

- **Sequence:** pelvic lymphadenectomy and anterior bladder mobilisation → **cystotomy away from the tumour** → excise tumour with full-thickness wall and perivesical fat, **1–2 cm mucosal margin**, confirmed by **frozen section** → excise ureteric orifice or intramural ureter and reimplant if needed → close with 2-0 polyglactin 910 in **2–3 layers**; fill via catheter to confirm watertight → **copious warm**

water irrigation (against pelvic seeding) → closed suction drain → **cystogram on day 7 before catheter removal**.

- **Perioperative intravesical chemotherapy** may be considered to reduce NMIBC recurrence.

Urachal Adenocarcinoma

- Rare primary adenocarcinoma, **mostly confined to the dome**, may extend directly.
- **En bloc excision of umbilicus, urachus and bladder dome** through a circumferential periumbilical incision extended toward the pubis, with visually clear margin confirmed by frozen section (Fig. 132.17); **bowel augmentation** if capacity is much reduced.

Concept — *Partial cystectomy applies radical-cystectomy oncology to a smaller specimen — same lymphadenectomy, same neoadjuvant chemotherapy, frozen-section margins — and is safe only where field disease (CIS, multifocality) is absent.*

ROBOTIC RADICAL CYSTECTOMY

Background

- Patients are difficult: pulmonary and cardiac comorbidity, **frailty, poor performance status**, advanced age, prior abdominal surgery or pelvic radiation.
- Enhanced recovery shortened stay and hastened bowel recovery, but **complications and transfusion rates remained stubbornly high** — the drive toward minimally invasive surgery.
- Evolution: **laparoscopic simple cystectomy → laparoscopic radical cystectomy with extracorporeal diversion → robotic radical cystectomy and lymphadenectomy → intracorporeal diversion** (first reported 2003; uptake rising per the International Robotic Cystectomy Consortium).
- Early resistance concerned oncological equipoise (no tactile feedback). **Unlike prostate cancer, bladder cancer has no effective salvage**: urine spillage, inadequate lymphadenectomy or a positive soft-tissue margin → **recurrence nearly always fatal**.

OPEN VERSUS ROBOTIC RADICAL CYSTECTOMY — THE TRIAL EVIDENCE

Study	Design / diversion	Primary endpoint	Verdict
Early randomised non-inferiority trial	Open diversion both arms	Node yield	No difference ; robotic: less blood loss, faster bowel return, less morphine; longer operation
Robotic extended dissection with open second look (MD Anderson)	Single arm	Adequacy of robotic node dissection	High robotic yield, few extra nodes, no stage change ; no positive margins in either early study
Memorial Sloan Kettering single-institution randomised trial	Open diversion both arms	90-day grade 2–5 complications (modified Clavien)	Closed for futility — complications similar ; less blood loss, higher cost, longer operation; later no difference in recurrence-free or cancer-specific survival
RAZOR (landmark, Lancet 2018)	Phase 3 non-inferiority; open diversion both arms	2-year progression-free survival	Robotic non-inferior (72.3% vs 71.6%) ; less blood loss, fewer transfusions, shorter stay
iROC (landmark)	Robotic with intracorporeal diversion vs open	Days alive and out of hospital at 90 days	Robotic better ; overall complications similar; fewer VTE and wound complications

Study	Design / diversion	Primary endpoint	Verdict
Meta-analysis, intracorporeal vs extracorporeal diversion	Expert centres	Complications	No difference overall; fewer major complications, less blood loss intracorporeally

- Open diversion in both arms is argued to affect a complication endpoint but **not progression-free survival**.
- Adoption has not matched robotic prostatectomy; demonstrated equipoise will raise interest. **Authors' verdict: oncological equivalence with recovery advantages; no compelling evidence of superiority of either approach.**

Patient Selection

- **Indications and staging are identical to open surgery.**
- Steep Trendelenburg and pneumoperitoneum → **raised pulmonary pressures, ventilation mismatch, raised intraocular pressure, reduced cardiac output** — address with anaesthesia before the day of surgery.
- **Prior abdominal or pelvic surgery is not a strict contraindication** (though excluded from trials); **avoid early in the learning curve after prior urological surgery** (partial cystectomy, ureteric reimplantation) — positive-margin risk.
- **Obesity:** positioning difficulty and worse ventilation in Trendelenburg; team decision; **when in doubt, open or convert.**
- Technique described for the **da Vinci system (Intuitive Surgical)**; principles transfer to new platforms.

Patient Positioning

- Tailored case by case: prior radiation, rectal or bowel risk, sexual and endocrine preservation, limited limb movement, existing prosthetic reservoirs.

DOCKING BY PLATFORM

	da Vinci Xi (rotating overhead boom)	da Vinci S, Si, X (no rotating boom)
Docking	Side-docking from the patient's right , men and women	Between the legs, lithotomy — side docking causes arm clashes
Position	Men usually supine ; lithotomy if vaginal access needed or rectal complications likely (prior prostatectomy or prostate radiation) — still dock right, leaving the space between the legs free for an assistant	Lithotomy

- **Anti-migration:** **3M Reston adhesive foam** under the torso (grips bed, cushions skin); arms tucked; padded chest strap at mid-chest; padded leg straps if supine (Fig. 132.18).

Port and Assistant Placement

- Ports sit **more cephalad than for prostatectomy** — extended lymphadenectomy, and right ports higher for terminal ileum and caecum if robotic diversion is planned (Fig. 132.19).
- **Access:** Veress needle, meniscal drop test, **insufflate to 15 mmHg** → camera port; inspect with patient flat before Trendelenburg → remaining ports.

PORT POSITIONS

Port	Position (text)	Figure legend variant
Camera	3–4 fingerbreadths (5–7 cm) above umbilicus	Same

Port	Position (text)	Figure legend variant
Right robotic × 2	Level with camera, straight line, 9–10 cm apart	Same
Left robotic	Nearer pelvis, level with umbilicus, ~15 cm from camera	14–16 cm lateral to umbilicus, just above it
12-mm assistant (or 5-mm)	Between left robotic port and camera	9–10 cm lateral to, level with, camera
15-mm assistant	Just cephalad to anterior superior iliac spine	9 cm lateral to left robotic port, just below umbilicus

- The **15-mm port** gives the best angle for bowel stapling, **removes large node packets en bloc** and admits a large Endo Catch retrieval bag.
- Port-closure sutures for 12- and 15-mm sites placed **at the start** (0 polyglactin 910, Carter-Thomason CloseSure).
- **Single assistant on the patient's left; four-arm configuration with two arms on the right; ~25° Trendelenburg**, then dock; insufflation may be reduced.

Instrumentation

TABLE 132.1 ROBOTIC SYSTEM RECOMMENDATIONS

Arm (patient left → right)	Instrument	Notes
1	Maryland bipolar forceps	Principal dissector (left medial port); second Cadiere forceps here during diversion
2	Camera, 0° lens	Nearly all cases, including low retroperitoneal (paracaval, para-aortic, preaortic) dissection; 30° lens rarely needed
3	HotShears (monopolar curved scissors)	Principal cutting instrument (right medial port)
4	Cadiere forceps	Retraction with less crush injury than ProGrasp ; limited bowel grasping, with care, under direct vision
1 and 3	Large needle drivers	Optional large SutureCut driver — especially for continent diversion

- The pattern is **bipolar left hand, monopolar right hand, gentle grasper in the fourth arm**; the trap is using ProGrasp on bowel.

Sigmoid Release and Posterior Dissection

- **Sequence:** grasp sigmoid **distal to the S-curve and retract cephalad (not to the right)** → release the peritoneal attachments forming the S-curve → sigmoid self-retracts out of the pelvis, freeing the fourth arm → continue along pelvic sidewall and **left white line of Toldt** (Fig. 132.20A–C) → incise peritoneum **just medial to the left ureter** → avascular plane beneath sigmoid mesentery, over presacral packet (Fig. 132.20D–E) → continue to the **right common iliac artery** (Fig. 132.20F).
- **Bleeding here means the sigmoid mesentery has been entered** → **adjust the plane; mesenteric fat is darker yellow than retroperitoneal or lymphatic fat**.
- Male cul-de-sac: peritoneum incised, dissection to the **prostatic apex** as in prostatectomy (Fig. 132.21); **seminal vesicles released only if nerve sparing is planned**.

Lateral Space of Retzius and Anterior Vascular Pedicle Dissection

- Women having oophorectomy: **ovarian vessels controlled now — two clips proximally, one distally** (large Weck clips).
- **Sequence (Retzius):** medial umbilical ligament (obliterated umbilical artery) grasped and retracted medially → bladder–abdominal wall interface exposed (Fig. 132.22A–B) → peritoneum incised to internal ring; **vas deferens / round ligament controlled** (bipolar or clips) → keep perivesical fat on bladder medially and nodes laterally → distally to **endopelvic fascia** (Fig. 132.22C), posteriorly to **hypogastric artery** (Fig. 132.22D) → in men endopelvic fascia incised as for prostatectomy (high release if nerve sparing).
- **Sequence (anterior pedicle):** **Weck clips preferred** (secure, no heat spread, nerve-sparing compatible) → upward and medial traction on obliterated umbilical artery opens the space to the ureter → **avascular plane medial to the pedicle** (Fig. 132.23A–B) → clip sequentially toward endopelvic fascia → **a broad distal hypogastric venous branch just proximal to the endopelvic fascia marks a complete anterior pedicle dissection** (Fig. 132.23C–D).
- **Ureter:** lift bladder near detrusor hiatus to elevate it (Fig. 132.23E) → blunt retraction with smooth instrument surfaces, bipolar to perforators → **never grasp the ureter** → mobilise above the common iliac artery (Fig. 132.23F) with **ample adventitia (prevents ischaemic stricture)**; no vessel loops needed.
- **Do not divide the ureters yet — complete the lymphadenectomy with antegrade urine flow;** right side then dissected similarly.

Extended Pelvic Lymphadenectomy

- **Why lymphadenectomy before cystectomy (ureters in continuity):** avoids prolonged ureteric occlusion (renal function), **keeps urine output as a resuscitation guide** for anaesthesia; occlusion-induced dilatation offers no reliable anastomotic advantage; ureters do not obstruct the field.
- Landmarks are those of open surgery; the **camera angle changes the visual cues**.
- Keep each packet **whole and bag it immediately**; the 15-mm port avoids fragmenting packets and transecting nodes.

ROBOTIC NODE PACKETS — MANOEUVRE, COMPLETION LANDMARK AND TRAP

Packet	Key manoeuvre	Completion landmark	Trap
External iliac (Fig. 132.24)	Lateral retraction of spermatic cord shows lateral border; tissue between artery and vein included; separated from obturator packet along natural cleavage plane	Distally to inferior epigastric vessels	Cutting across without cord retraction transects nodes; genitofemoral nerve branches (femoral, inguinal) variable — release, or en bloc if densely adherent in advanced disease
Obturator (Fig. 132.25)	Suction-tip retraction of external iliac artery; release from vein and sidewall; bipolar to lateral perforators	Lateral and posterior release until hypogastric vein wall is seen; fossa bare	Obturator nerve is not the posterior limit ; dissecting at the nerve transects nodes
Hypogastric (internal iliac) (Fig. 132.26)	Deep to obturator nerve; release from sidewall toward endopelvic fascia; gentle upward traction	Deep extension lateral to mesorectal fascia removed	Leaving the deep posterior portion
Common iliac (Fig. 132.27)	From lateral border of common iliac artery, posteriorly	Common iliac vein wall seen; ureter lies over cleared bed	—

Packet	Key manoeuvre	Completion landmark	Trap
Presacral (Fig. 132.28)	Incise thinned peritoneum over right common iliac artery to join left-sided dissection; bipolar to perforators	Aortic bifurcation seen; packet lifted off left common iliac vein and presacral space	May be omitted in NMIBC with nerve sparing to protect hypogastric plexus

- Extent is tailored to cancer characteristics and functional goals — **extend into the retroperitoneum or omit portions.**

Ureteral Control and Posterior Pedicle Dissection

- **Sequence:** ureters **clipped at the detrusor hiatus** (more proximally if trigone or intramural ureter involved) (Fig. 132.29A–B) → space between clips → **ureteric margin frozen section** (Fig. 132.29C) → **left ureter transposed beneath sigmoid mesentery to the right lower quadrant** (Fig. 132.29D); tether sutures unnecessary if sigmoid release adequate → **posterior pedicle controlled before urachus and anterior Retzius release** (better view) → men: rightmost arm retracts bladder upward toward opposite shoulder (Fig. 132.30A); vessel sealer (**clips if nerve sparing**) toward genitourinary diaphragm (Fig. 132.30B) → prostatic pedicle (Fig. 132.30C) → prostatic apex (Fig. 132.30D).

Male Apical Dissection

- **Sequence:** urachal attachments released (bipolar) → **dorsal venous complex: 2-0 V-Loc figure-of-eight, suspended through the pubic symphysis synchondrosis** (Fig. 132.31A) → urethra isolated (Fig. 132.31B) → catheter removed; urethra controlled proximally and distally and transected; **neobladder planned → proximal clip only, no distal clip** (Fig. 132.31C) → **distal urethral margin:** partially transect close to the proximal clip, then complete a few millimetres distally (Fig. 132.31D) — easier than after specimen release.

Female Radical Cystectomy

- **En bloc with uterus and ovaries as in open surgery.** An **EEA sizer placed vaginally** identifies the posterior cervical fornix (or vaginal vault after hysterectomy), gives countertraction and distends the wall for the sparing plane.
- **En bloc anterior vaginectomy:** posterior fornix opened → vessel sealer along anterolateral vaginal wall to introitus → at meatus, **catheter detached from bag, clipped by assistant and pulled intracorporeally** to retract the meatus → complete urethral and meatal excision (Fig. 132.32E–F).
- **Vaginal sparing (uterus in situ; Fig. 132.32):** fourth arm lifts bladder, EEA countertraction (A) → peritoneum incised, anterior vaginal wall exposed (B) → midline dissection toward introitus in a **near-avascular plane** → lateral posterior pedicles defined as in men (C) → clips or sealer to the meatus (D) → neobladder: apical dissection after urachal release; no urethral sparing; open vagina just behind bladder neck with the catheter manoeuvre.
- **Vaginal closure (clamshell):** authors use **running 2-0 V-Loc** (alternatives interrupted or figure-of-eight 2-0 polyglactin 910); **mucosa-to-mucosa, watertight** — otherwise **peritoneal fistula and prolonged vaginal leakage; Premarin-coated pack, removed day 2.**
- **Specimen extraction:** after en bloc anterior vaginectomy, bagged specimen delivered through the vaginal defect under tension monitoring; **abandon if the bag stalls or a tear threatens** → extend the midline port site.

Postoperative Care

- Genitourinary and gastrointestinal surgery in elderly, comorbid patients: **major complications are infrequent but most patients have some complication — commonest gastrointestinal, then infectious.**
- Monitor blood count, electrolytes, renal function and fluid balance; often **intensive care or step-down** initially.
- **No routine nasogastric tube** — only for impaired mentation or airway-protection concerns.
- **Continue VTE prophylaxis unless bleeding; early ambulation;** incentive spirometry, deep breathing, coughing.
- **Delayed bowel function prolongs stay: alvimopan; neostigmine (with telemetry);** promotility suppositories; stress ulcer prophylaxis; antiemetics; **early enteral feeding** if no nausea or vomiting; **opioid avoidance — ketorolac and paracetamol** unless contraindicated.

Concept — *Robotic cystectomy is judged on oncology first because bladder cancer has no salvage; randomised evidence shows equivalent progression-free survival with recovery gains, and each technical rule (whole node packets, never grasp the ureter, lymphadenectomy before ureteric division) protects the margin, the node count or the ureter.*

KEY POINTS

- Bladder cancer is common, costly and frequently fatal.
- Radical cystectomy offers excellent survival for organ-confined disease.
- TUR is diagnostic and therapeutic, especially for low-grade tumours.
- Perioperative intravesical chemotherapy reduces recurrence by about 13%.
- Standard template: genitofemoral nerve laterally, internal iliac artery medially, Cooper ligament distally, ureter crossing the common iliac artery proximally.
- Ureteric margin frozen section predicts upper tract recurrence, but sequential resection to a negative margin may have no impact.
- Nerve-sparing radical cystectomy aims to preserve sexual function and mirrors radical prostatectomy technique.
- Subtotal prostatic resection may further aid sexual function, only where occult prostate cancer risk is low.
- Vaginal sparing is appropriate without locally advanced disease and permits orthotopic diversion.
- Partial cystectomy suits rare, solitary urothelial tumours without concomitant CIS.
- Urachal adenocarcinoma: en bloc resection of umbilicus and bladder dome.
- Robotic radical cystectomy gives similar oncological outcomes to open and can speed recovery.
- Enhanced recovery protocols restore physiology sooner and shorten stay.